

Tasneem Jahan Farheen

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RESEARCH INTERESTS

My research focuses on developing trustworthy, agentic AI for clinical decision support. I aim to design multi-agent retrieval-augmented generation (RAG) frameworks that integrate multimodal healthcare data including EHRs, medical imaging, and clinical notes across diverse populations. My work prioritizes uncertainty quantification and calibration to prevent miscalibrated, overconfident predictions in critical care settings. Additionally, my interest lies on the examination of algorithmic fairness to eliminate reliance on non-clinical proxy confounders, developing deployable, clinician-trusted tools that advance diagnostic precision and equitable care.

EDUCATION

BRAC University , Dhaka, Bangladesh Master of Science in Computer Science and Engineering Thesis Title: Knowledge Distillation in Split Learning for Heterogeneous Clinical Tabular Data	May 2024 — March 2026 Cumulative GPA: 3.17/4.00
American International University-Bangladesh , Dhaka, Bangladesh Bachelor of Science in Computer Science and Engineering Thesis Title: Heart Disease Prediction Using Machine Learning	September 2019 — May 2023 Cumulative GPA: 3.93/4.00

ACADEMIC EXPERIENCE

Mastermind English Medium School <i>Computer Science Teacher</i>	Dhaka, Bangladesh January 2024 — Present
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- Teach Computer Science and Python coding to middle school students, covering basic programming logic and hands-on mini-projects.
- Introduce basic AI concepts so students understand how smart systems work and how to use them responsibly.
- Train students in Microsoft Excel to make tables, use basic formulas and create simple charts.
- Design and evaluate digital assessments for academic progress tracking.
- Conduct weekly lab sessions, improving hands-on skills by 85% based on assessments.

PROJECTS

Few-Shot Autism Screening via Dual-Mechanism Knowledge Distillation <i>Lead Researcher</i>	Dhaka, Bangladesh May 2026 — Present
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- Designed a split-learning knowledge distillation framework for cross-domain autism screening across Adult, Child and Adolescent tabular cohorts.
- Formulated two distillation paradigms: Platt-calibrated logit-level distillation and confidence-weighted encoder-level representation distillation.
- Evaluated three student architectures (TabM, MLP, SAINT-style) under strict few-shot settings.

Knowledge Distillation in Split Learning for Heterogeneous Clinical Tabular Data <i>Graduate Researcher</i>	Dhaka, Bangladesh May 2025 — March 2026
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- Proposed split learning framework for diabetes risk prediction across heterogeneous clinical datasets.
- Designed and evaluated logit-level and encoder-level distillation strategies under extreme few-shot supervision.
- Benchmarked MLP, SAINT and TabM student models demonstrating consistent gains over scratch training in low-resource settings.

Heart Disease Prediction Using Machine Learning <i>Undergraduate Researcher</i>	Dhaka, Bangladesh September 2022 — May 2023
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- Built and evaluated five machine learning models (Logistic Regression, Random Forest, SVM, KNN, and Naïve Bayes) for early heart disease detection.
- Cleaned clinical tabular datasets, handled missing values, and performed feature scaling and encoding.
- Applied 10-fold cross-validation and hyperparameter tuning, evaluating models on ROC-AUC, accuracy, and recall.

PUBLICATIONS

Journal paper

In-Progress

- **Farheen, T. J., & Alam, M. G. R.** (2026). Few-Shot Autism Questionnaire-Based Screening via Dual-Mechanism Knowledge Distillation.

Master's Thesis

Published

- **Farheen, T. J.** (2026). *Knowledge Distillation in Split Learning for Heterogeneous Clinical Tabular Data*. Master's thesis, Department of Computer Science and Engineering, BRAC University. Available at: BRACU Institutional Repository.

AWARDS

Merit-Based Graduate Scholarship (40%)

Awarded by BRAC University for postgraduate studies based on undergraduate academic excellence

Dhaka, Bangladesh

May 2024

Dean's List Honor

Faculty of Science and Technology, AIUB (Awarded for 3 semesters)

Dhaka, Bangladesh

2019 — 2023

OTHER EXPERIENCES

Upwork Inc.

Full Stack Developer

Remote

Oct 2022 — May 2023

- Built and maintained a Laravel-based PHP web application with a Vue.js front end following MVC architecture.
- Optimized core application code and database queries, boosting application performance by 20%.
- Managed collaborative version control using Git and supported end-to-end client deployment pipelines.

SKILLS

- **Programming Languages & Frameworks:** Python, PyTorch, Scikit-Learn, SQL, Java, PHP (Laravel), JavaScript (Vue.js, React)
- **Software & Tools:** Git/GitHub, VS Code, Google Colab, Microsoft Excel, Power BI
- **Research & Core Competencies:** Split Learning, Knowledge Distillation, Model Calibration, Data Preprocessing & Feature Engineering
- **Soft Skills:** Academic Writing, Curriculum Delivery, Technical Communication, Problem-Solving, Team Collaboration

ENGLISH LANGUAGE PROFICIENCY

IELTS Academic: Overall Band 7.0 (CEFR Level: C1)

Listening: 8.0 | Reading: 7.0 | Speaking: 7.5 | Writing: 6.0

Test Date: March 15, 2025

REFERENCES

Dr. Md. Golam Rabiul Alam

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BRAC University, Dhaka, Bangladesh

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Scholar Profiles: BRAC University - Personal Page — Google Scholar — LinkedIn

Tanvir Ahmed

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